Java Threads – Practice Questions

1. **Thread Life Cycle**
   Write a Java program that demonstrates the different stages of a thread's life cycle. Create a thread, start it, make it sleep for a specified period, and allow it to complete. Explain the transition between the **New, Runnable, Running, Waiting/Timed Waiting, and Terminated** states.

2. **Creating Threads using Thread Class**
   Write a Java program to create **three threads by extending the Thread class**. Each thread should perform a different task, such as printing numbers, displaying characters, and displaying a message. Execute all three threads concurrently and observe their execution order.

3. **Creating Threads using Runnable Interface**
   Write a Java program to create **two or more threads by implementing the Runnable interface**. Each thread should perform a separate task. Explain why implementing Runnable can be preferable to extending the Thread class in certain situations.

4. **Synchronization**
   Write a Java program in which multiple threads access and update a **shared bank account balance**. Demonstrate the problem that can occur when multiple threads modify the balance simultaneously. Then use the **synchronized keyword** to ensure that the balance is updated correctly.

5. **Multithreading with Synchronization**
   Develop a Java program that simulates a **ticket booking system**. Create multiple threads representing customers attempting to book tickets from the same limited ticket pool. Use synchronization to ensure that two customers cannot book the same ticket. Display the booking details and remaining number of tickets after each transaction.